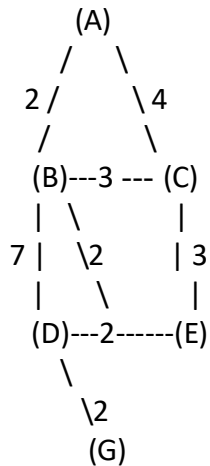


PROBLEMS

Question 1: Dry Run of A Search Algorithm (10 Marks)*

Consider the following graph.



Heuristic Values

Node $h(n)$

A	7
B	6
C	4
D	3
E	2
G	0

Task:

Perform a **dry run of the A* Search algorithm** from **A** to **G**.

For each iteration, show:

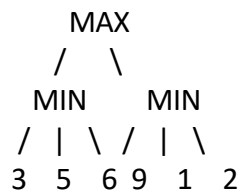
- Current node
- Open List
- Closed List
- $g(n)$
- $h(n)$
- $f(n) = g(n) + h(n)$

Finally,

1. Identify the optimal path.
2. Calculate the total path cost.

Question 2: Dry Run of Minimax with Alpha-Beta Pruning (10 Marks)

Consider the following game tree.



Task:

Perform a **dry run of the Minimax algorithm with Alpha-Beta Pruning** by evaluating the tree **from left to right**.

For each node, show:

- Alpha (α)
- Beta (β)
- Selected value
- Any node that gets pruned

Finally,

1. Determine the best move for MAX.
2. State the final minimax value.
3. List the pruned nodes (if any).

